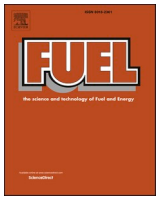
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Full Length Article



Catalytic hydroprocessing of waste cooking oil for the production of drop-in aviation fuel and optimization for improving jet biofuel quality in a fixed bed reactor

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ABSTRACT

Waste cooking oil (WCO) is a bio-waste with significant potential for developing improved transportation fuel, specifically for the aviation industry. This work investigates the catalytic hydrocracking of WCO over commercial hydrocracking catalyst (Ni-Mo/silica-alumina) in a continuous flow fixed bed reactor to produce jet range hydrocarbons (HCs). The aim was to convert WCO into sustainable aviation and automotive fuel to get the maximum benefit from waste oil with high selectivity towards paraffins, naphthenes, minimal aromatics, and high isomerization (isomer to normal (i/n) ratio) in a single-step process using a single catalyst composition. The higher i/n ratio boosts specific energy, thermal stability and provides a low freezing point to the fuel. The results show that WCO conversion of 96.6 % with 55.6 % selectivity of jet range HCs was obtained at 420 °C temperature, 1 h^{-1} LHSV, 10 MPa pressure, and 2200 NL_{gas}/L_{liquid} H_2/oil ratio. In addition, isomer yield of 82.9 %, i/n ratio of 4.8, and selectivity of kerosene range HCs based on paraffins (43.6 %), naphthenes (8.2 %), and aromatics (3.73 %) was obtained at optimum reaction conditions. After hydroprocessing, the major fraction of heteroatoms (N = 99.9 %, S = 94.3 %, O = 99.9 %) was removed from WCO. Finally, the time on stream (TOS) study showed no deactivation of the catalyst up to 590 h, indicating the long life and stability of the catalyst, which is one of the highest lifetimes reported so far. Bio-jet fuel properties were tested according to the ASTM D-1655 standard. This research will aid in developing novel catalysts for the selective conversion of WCO into jet range HCs in a single step process.

1. Introduction

The usage of petroleum-based jet fuels pollutes the environment as petroleum fuels are the major producer of greenhouse gases (GHG). Apart from these pollutants, petroleum-based fuels are the substantial generator of NO*x*, SO*x*, CO, particulate matter, and volatile organic compounds [1]. Biomass-derived aviation fuel has long been seen as a renewable energy choice as it minimizes the usage of arable land, GHG radiation and reliance on fossil fuels. Renewable jet fuel has been taken into consideration as a potential renewable opportunity and long-term solution to relieve the approaching troubles in the aviation industry. From 2008 to 2025, aviation fuel global consumption is predicted to rise by nearly 38 %, with an annual enhancement rate of 1.9 % [2]. Biomass-derived aviation fuel can substitute about 30 % of total aviation fuel [3].

Furthermore, when compared to petroleum-based jet fuel, bio-jet fuel may lessen GHG emissions by 65–85 % [4]. For manufacturing biomass-derived aviation fuel, many paths have been explored, including catalytic sugar-to-jet process [5], gasification to Ficher Tropsch process [6], alcohol-to-jet process [7] and hydroprocessed renewable jet (HRJ) [8]. One of the most important methods for turning non-edible oil-based feeds into aviation fuel is HRJ. Several conversion methods have been developed and tested on a commercial or pre-commercial scale in recent years. The HRJ method has been regarded as the sole commercially feasible technique, and several test flights using HRJ fuel have been developed [9]. In the hydroprocessing process, triglycerides present in the feedstock transform into straight-chain alkanes

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via the hydrodeoxygenation route and these alkanes changes to light and branched alkanes via catalytic hydrocracking and hydroisomerization to meet jet fuel standards. For fulfilling the cold flow qualities while retaining the fuel's flash point, the HRJ fuel is typically made up of straight-chain alkanes and isomer alkanes with carbon numbers ranging from 9 to 15.

Apart from the necessity of creating and utilizing biofuels, there are various challenges with the current manufacturing procedures. The feedstock is an important and crucial part of the procedure. WCO is the oil remainder left over after cooking that has deposited free fatty acids allowing it a potentially effective and feasible biofuel feedstock [10]. WCO is two to three times less expensive than pure vegetable oils, significantly lowering the cost of manufacturing biofuels [11]. The conversion of WCO to biofuels also reduces the severe environmental effects of dumping spent oils into drainage systems [12]. Furthermore, the hydroconversion of triglycerides is a complex process (saturation of double bonds, heteroatom removal, cracking, and isomerization) for which bi-functional catalysts are essential. Frequently, NiMo sulfide phase is employed for the hydrocracking process, due to its high hydrogenation and hydrodesulfurization activity [13–15]. The support must have a high surface area and large pore size to allow the bulk molecules to access the active sites located inside the pores easily. The silica and alumina are frequently utilized supports in this kind of process due to their low cost, excellent thermal and hydrothermal stability. Sulfided hydrocracking catalysts are obviously suitable options for deoxygenating triglycerides in non-edible oils to biofuels. However, relatively little is known about their activity and selectivity towards jet range hydrocarbons from WCO.

Bezergianni et al. have carried out numerous experiments for catalytic hydroprocessing of WCO into high-quality fuels through one (hydrotreating) and two-step (hydrotreating + isomerization) methods [16]. Hydroprocessing of WCO has been studied utilizing three catalysts; a hydrotreating, a mild and a severe hydrocracking catalyst from temperature 330 to 390 °C and 8.27 to 13.79 MPa pressure under fixed LHSV (1 h^{-1}) [17]. The results indicate that hydrotreating catalyst exhibited the highest conversion (83 %), and diesel yields were over 94 % in all cases. The temperature was identified as the most important factor in transforming WCO into biofuels [18]. The influence of hydrotreating pressure, LHSV and H_2/oil ratio on biofuel generation was also investigated [19]. The reaction route for synthesizing WCO to biojet fuel has been suggested by Li et al. The products contained 53 % jet range alkanes and 13.4 % aromatic HCs [4]. Cheng et al. used soybean oil as a feedstock and converted it into jet fuel with 40.3 % alkanes and 23.8 % aromatics HCs [20]. Another study by Liu et al. in which they have developed HRJ fuel by using castor oil, yielding 91.6 wt% aviation range alkanes and an i/n ratio of 4.4–7.2 [21]. Hydroconversion of WCO into biofuels was reported with 39.7 % jet fuel selectivity and 18.7 % jet range aromatics [22]. Nevertheless, a minimum concentration of aromatics must be present in the HRJ fuel in order to reduce soot (carbon particles) production from the airplane engine [23]. There have been few kinds of research conducted to evaluate the impact of catalyst and operating parameters on product distribution in the final products. Efficient catalysts with longer lifetime and stability towards HRJ production from hydroprocessing of WCO in a single-step process using a single catalyst remain under-explored and need to be studied further to produce HRJ fuel with significant concentrations of paraffins, naphthenes and minimum amount of aromatics as well as high isomers formation to improve the jet fuel quality.

In light of this, the present study focuses on the hydroprocessing of WCO in a fixed-bed down flow reactor over Ni-Mo/silica-alumina catalyst to look into its suitability to produce bio-aviation fuel. The products with high concentrations of paraffins, cycloparaffins, low aromatics and high isomerization (i/n ratio) are the primary target to meet the jet fuel specifications. Such information is necessary for hydrocracking catalysts to be further optimized in order to become effective deoxygenation and selective cracking catalysts with high stability and a longer lifetime. No reports are available for the hydrocracking process to produce Jet A-1 aviation fuel from inedible WCO in a single-step process with high isomerization, minimum aromatics, and longer lifetime and stability of the catalyst, which makes this work new and important.

2. Methodology

2.1. Materials

The feedstock utilized in this study was 100 % WCO and collected from local hotels, households and restaurants of Dehradun, India, after being widely used for frying. The WCO was filtered through a standard sieve before being introduced to the experimental area to eliminate any food material that remained in the oil after cooking/frying. Table 1 lists the typical chemical properties of WCO. For this study, a commercial Ni-Mo/silica-alumina catalyst was employed for the catalytic hydroprocessing of WCO in all the experiments. The catalyst is presulfided, as described in Section 2.3.

2.2. Experimental setup

In this study, all hydrocracking experimental tests for converting WCO to kerosene range HCs were carried out in a fixed bed down flow microreactor with a commercial hydrocracking catalyst. This type of microreactor can be employed for catalytic hydroprocessing reactions at high temperatures and pressures. The microreactor can be characterized into four sections: hydrogen section, feed insertion section, reactor section and product collection/separation section, as depicted in Fig. 1. A back pressure regulator (BPR, Tescom) and a mass flow controller (MFC, Brooks) were used to properly manage the hydrogen pressure and gas flow of the reactor. The reactor was heated by an electric furnace (Serial no 10-5887-1-2). The interior and exterior temperatures of the reactor were monitored with Type K thermocouples calibrated to 1 K accuracy and controlled with a temperature controller. A high-pressure liquid metering pump (Eldex) was used in the system to regulate the liquid flow of the feed properly. Uniform contact of the feedstock with catalyst and hydrogen causes a hydrotreating/hydrocracking process at high temperature and hydrogen pressure in the fixed bed reactor. After direct contact of feedstock and catalyst within the reactor, the product reaches a high-pressure separator (HPS) in the form of a mixed gas–liquid phase. HPS was used to remove the gaseous components, including hydrogen and some other lighter gases, such as propane, carbon dioxide, and carbon mono oxide, from the hydrocracked

Table 1
Chemical and physical properties of WCO.

Properties	Value
Density (g cm^{-3} at 15 °C)	0.9251
Kinematic Viscosity (mm^2 s^{-1} at 40 °C)	47.77
Kinematic Viscosity (mm^2 s^{-1} at 100 °C)	9.87
Moisture content (%)	0.164
Acid value (mg KOH/g)	2.32
Free fatty acid	1.16
N (ppm)	320
S (ppm)	390
C (wt%)	76.82
H (wt%)	11.46
O (wt%)	11.649
Na (ppmw)	4.24
K (ppmw)	1.2
Ca (ppmw)	2.32
Mg (ppmw)	7.28
Fe (ppmw)	0.5
C16:0 (Palmitic acid)	17.1
C16:1 (Palmitoleic acid)	3.5
C18:0 (Stearic acid)	5.9
C18:1 (Oleic acid)	37.8
C18:2 (Linoleic acid)	35.6

Pressure	0-200 bar
Temperature	0-550 °C
Liquid flow rate	0.02-2.5 ml/min
H_2 flow rate	0-1500 ml/min
Catalyst volume	1-6 ml
Reactor i.d.	8 mm
Reactor length	455 mm

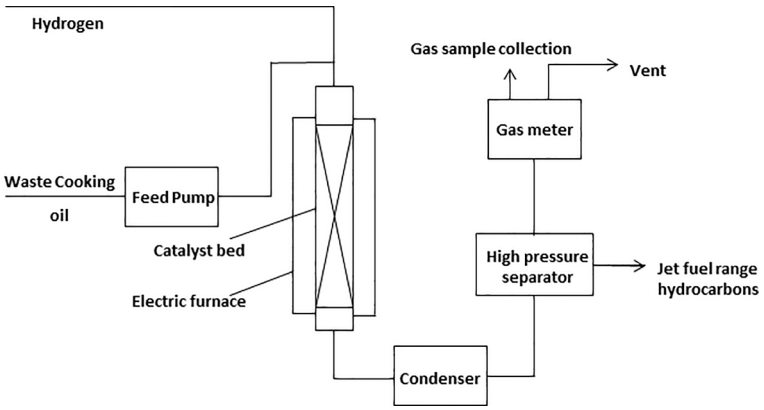


Fig. 1. Simplified schematic diagram of microreactor unit employed for hydroprocessing of WCO.

product. The gas was discharged using a drum-type gas meter (Ritter, 0.54G.916) and examined. The hydrocracked products of the catalytic reaction were collected and studied on a daily basis to investigate the influence of temperature, LHSV, hydrogen pressure, and H_2/oil ratio.

2.3. Experimental procedure

To maintain the required bed length of the catalyst and improve the thermal resistance of the reaction, a non-sulphided catalyst (4 g or 5.5 ml) was blended with SiC (5.5 ml) and introduced into the microreactor (45.5 cm length). The catalyst was put in the centre of the reactor to ensure adequate heating and flow stability. The catalyst utilized in this study is presulfided using 2.5 wt% of Dimethyl Disulphide (DMDS) in straight run gas oil (SRGO) to retain the catalytic activity as indicated by the catalyst supplier. Before carrying out the experiments, the catalyst was dried in the reactor at 175 °C for 5 h. Then after, SRGO was used to sulfide the catalyst at 250 °C, 300 °C, 320 °C, and 350 °C for 5, 5, 6 and 2 h, respectively.

The sulfiding agent DMDS (0.025 wt%) was utilized in combination with feed to limit the sulfur reduction from the catalyst. Firstly, a waiting period of 12 h (first period) was set out for the reaction process to stabilize before collecting the final products (second period). A GC analysis of the products collected at both periods was performed to confirm the persistent catalytic activity and ensure that the catalyst was not deactivated. Products were categorized into three groups based on their carbon number; <C9 carbon range as naphtha, C9 - C15 as kerosene, and C16 - C18 as diesel. Individual products were identified using GC standards.

A detailed study was used to calculate the conversion based on the mole balance of total liquid products. The triglyceride concentration converted to any form of intermediates or HCs is known as triglyceride conversion. The mole balance was 96–99 % precise for all operating conditions before and at the end of the reaction. WCO conversion was determined using the following formula:

$$Conversion\ (\%) = \frac{C_{feed} - C'_{product}}{C_{feed}} \times 100 \quad (1)$$

where C is the total amount of triglycerides (TG) present in the feed (WCO) and C' is the total amount of triglycerides in the final product.

Furthermore, the measure of selectivity is used to investigate the efficacy of hydrocracking for the formation of a certain product rather than other products. The selectivity of naphtha, kerosene and diesel range HCs was calculated according to the following equation [18] to analyze the effectiveness of the hydroprocessing reactions.

$$\text{Product selectivity}\ (\%) = \frac{\text{Product}_{a-b} - \text{Feed}_{a-b}}{\text{Feed}_{331+} - \text{Product}_{331+}} \times 100 \quad (2)$$

where $Feed_{331+}$ and $Product_{331+}$ are the weight percent of the heavy molecules (>C18+) present in the feed and corresponding product, with a boiling point higher than 331 °C. $Feed_{(a-b)}$ and $Product_{(a-b)}$ are measured as the wt% of the feed and product in which 'a' and 'b' are the initial and final boiling points for a particular HC range. Since 'a' and 'b' boiling range is different for various range of carbon number, hence, selectivity can be calculated for naphtha (68–130 °C), kerosene (130–260 °C), and diesel (260–331 °C) using the equation (2).

Also, hydroisomerization effectiveness in the final products was measured by calculating the isomer yield and i/n ratio. The isomer yield is calculated as follows:

$$\text{Isomer yield}\ (\%) = \frac{\text{i}}{\text{i} + \text{n}} \times 100 \quad (3)$$

where i and (i + n) signify the quantity of isoalkanes and the total quantity of normal and iso-alkanes, respectively, in the final product of kerosene range HCs (C9 - C15).

The i/n ratio is the ratio of the total amount of iso-alkanes to normal alkanes in the final product of kerosene range HCs.

2.4. Product analysis

The obtained products after the hydrocracking reactions were distilled into naphtha, kerosene and diesel range HCs using a 9600 High-Efficiency Distillation Unit (B/R instruments, USA). The simulated distillation curve was determined via Varian 3800-GC according to the ASTM-2887-D86 method. Rudolph research analytical equipment (DDM-2911) was employed for measuring the density of feed and final liquid products. A residual gas analyzer (RGA) from Agilent 7890A with two TCD detectors was used to analyze the gases. Elementar Micro-cube CHNS analyzer was employed for determining C, H, N and S percentages in the feedstock and products. The liquid products and unconverted WCO obtained after the hydrocracking reactions (isomerization, deoxygenation, cracking) were analyzed after separating the water phase by gas chromatography (GC) analysis (Agilent Technologies GC (7890B, G3440B)) fitted with a flame ionization detector (FID) and thermal conductivity detector (TCD). Vf-5 ms column (30 m × 0.25 mm, 0.25 um) was used for quantification. At first, the column temperature was maintained at 40 °C with a holding period of 0.5 min; after that increased to 70 °C, 150 °C, 200 °C, 350 °C with the rate of 3 °C, 5 °C, 15 °C, 20 °C per min, and lastly 370 °C with the rate of 30 °C per min having 12 min holding time. Errors in the experiments were also identified and accounted for in the final results. All the ASTM methods were applied in accordance with the "Standard Specification for Aviation Turbine Fuel containing Synthesized Hydrocarbons" [24], and the aviation kerosene was tested particularly for ASTM D7566.

3. Results and discussion

A set of experiments were performed to check the influence of hydrocracking temperature, LHSV, pressure and H_2/oil ratio on the catalytic hydroprocessing of WCO into biofuels. In addition, a catalyst lifetime (TOS) study was also investigated.

3.1. Influence of hydrocracking temperature on product distribution

The temperature has been recognized as a critical factor for catalyst efficacy and life [25]. The rise in the temperature generates an increase in the catalyst activity, which causes the catalyst life to degrade quicker. As a result, the influence of reactor temperature on types of reactions such as hydrocracking, hydroisomerization and heteroatom removal was investigated. The hydroprocessing of WCO is a complicated process in which various chemical reactions occur sequentially or simultaneously, including hydrogenation/dehydrogenation, deoxygenation, hydroisomerization, and hydrocracking. All reactions were carried out on WCO hydrocracking at five different temperatures, i.e. 360, 380, 400, 420, and 440 °C. The rest operating parameters were; P = 10 MPa, LHSV = 1 h^{-1}, and H_2/oil ratio = 2200 NL_{gas}/L_{liquid}.

The temperature of the hydrocracking process has a significant impact on product yields. The overall liquid product yields were calculated using simulated distillation data at each reactor temperature. The naphtha, kerosene and diesel yields were determined as the wt% of the product having boiling points ranging from 68 to 130 °C, 130 to 260 °C and 260 to 331 °C, respectively. After the hydrocracking reactions, lighter products (gaseous components) and unconverted

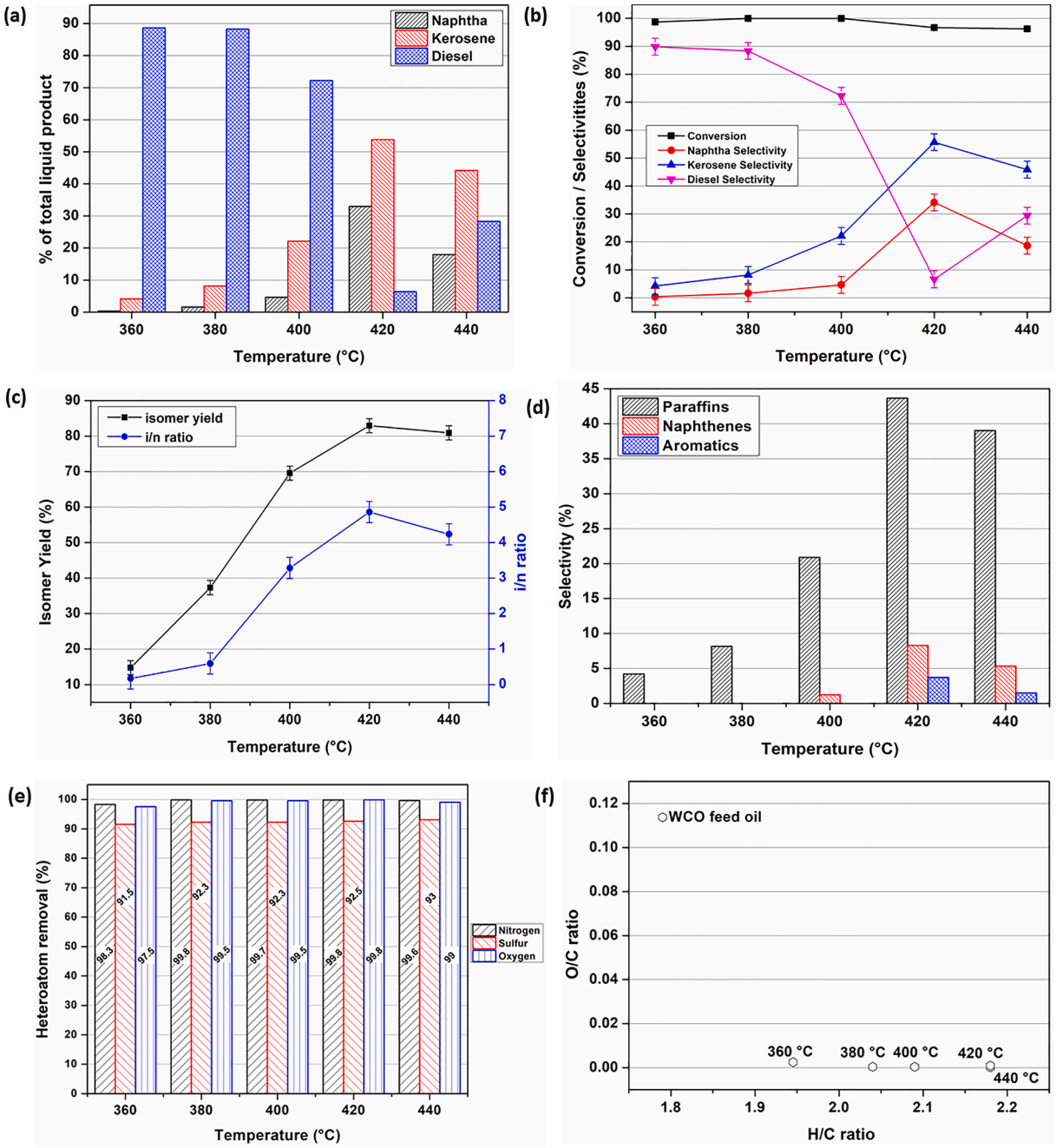


Fig. 2. Influence of hydrocracking temperature on (a) naphtha, kerosene and diesel product yields (wt%), (b) conversion and selectivities, (c) isomer yield and i/n ratio, (d) PNA%, (e) heteroatom (nitrogen, sulfur, oxygen) removal percentage, (f) molar (O/C) and (H/C) ratio by Van Krevelen diagram for feed and hydrocracked products. Constant operating conditions were; LHSV = 1 h^{-1}, P = 10 MPa, H_2/oil ratio = 2200 NL_{gas}/L_{liquid}.

portions with a boiling point of less than 68 °C and greater than 331 °C, respectively, cannot be utilized as liquid biofuels.

The comparison of naphtha, kerosene and diesel product yields resulting from the hydrocracking of WCO is shown in Fig. 2(a). Diesel and kerosene yield was dominant at lower and higher temperatures, respectively. In particular, diesel had the maximum yield of 88.6 % at the lower hydrocracking temperature (360 °C). However, as the temperature rises, the yield of diesel drops because higher temperatures favour cracking and, as a result, the generation of kerosene and naphtha molecules. Notably, the kerosene yield increased from 4.1 % at 360 °C to 53.8 % at 420 °C, while naphtha yield increased from 0.3 % (360 °C) to 32.9 % (420 °C). The maximum kerosene yield was observed at 420 °C of the total liquid product. Therefore, a smaller temperature should be considered for diesel and a higher temperature for kerosene production.

The conversion and product selectivities (naphtha, kerosene and diesel) at five various temperatures are illustrated in Fig. 2(b). The conversion and selectivities were estimated using equations (1) and (2), respectively, using the simulated distillation data at each hydrocracking temperature. The conversion gradually increased from 98.6 to 99.9 % with a rise in the temperature from 360 to 400 °C because hydrogenolysis and hydrocracking activity are endothermic reactions that become more favoured with increased temperature. Increment in the temperature from 400 to 440 °C reduces the conversion from 99.9 to 96.2 % because the reaction phase shifted primarily to vapour, reducing the residence time while maintaining the high hydrogen flow [26]. As a result, the conversion rate was reduced due to insufficient reaction time [26]. Diesel selectivity was dominant (89.8 %) at the lowest temperature (360 °C). Further increase in temperature from 360 to 380 °C, diesel selectivity slightly reduced from 89.8 to 88.3 %, while kerosene selectivity raised from 4.2 to 8.1 %. But when the temperature was increased from 380 to 420 °C, diesel selectivity was reduced from 88.3 to 6.6 %, while the produced kerosene was increased from 8.1 to 55.6 %. Moreover, naphtha selectivity was also increased from 0.3 to 34.1 % when the temperature was increased from 360 to 420 °C. This was predicted since higher temperatures generated more severe cracking, resulting in the additional breaking of diesel HCs into kerosene and naphtha (lighter) range HCs. As a result, a higher hydrocracking temperature (420 °C) promotes kerosene selectivity, and lower hydrocracking temperatures (360 and 380 °C) are suited for diesel production. It should be noted that Bezergianni et al. [27] produced kerosene with 20.04 % selectivity from catalytic hydrocracking of WCO at 390 °C. However, 35.7 % of jet fuel selectivity was obtained by Wang et al. [22] using a zeolite-supported Ni-Mo catalyst. In another study by Wang et al. [28], bio-jet range HCs was synthesized from WCO with 37.3 % selectivity at 380 °C. It is evident from the research study [22,27,28] that the temperature needed for hydrocracking of WCO is lower than the present work. However, products obtained in this study have relatively high selectivity towards jet range HCs.

Fig. 2(c) shows the isomer yield and i/n ratio of the products obtained after the hydrocracking of WCO at various reactor temperatures. The quantity of isomer was lowest (14.7 %) at 360 °C temperature, and grew to 69.5 % by raising the temperature from 360 to 400 °C and reached the maximum of 82.9 % at 420 °C. This is understood by the fact that higher temperature accelerates the catalyst's activity, leading to increased isomers formation. The amount of isomer declined from 82.9 to 80.9 % when the reactor temperature increased above 420 °C, as a shorter reaction time at a higher temperature lowered the efficacy of hydroisomerization and hydrocracking. Furthermore, the i/n ratio was low (0.17) at 360 °C and increased with the temperature. The maximum i/n ratio (4.8) was obtained at 420 °C. It could be due to the fact increasing the reaction temperature to a reasonable level encourages the isomerization process. [29]. Hydroisomerization, on the other hand, is a weak exothermic process that may be stopped by high temperatures. As a result, at 440 °C, the number of isomers reduced, and the i/n ratio fell to 4.2. Wang et al. [22] produced jet fuel having an i/n ratio of 2.7 using core–shell zeolite supported Ni-Mo catalyst at 380 °C. Wang et al. [26,30] conducted a study in which they employed a two-step process to produce jet range HCs with i/n ratio of 0.7. Thus, the i/n ratio in this study was significantly higher than the one in [22,26]. It is obvious that the catalyst used in this study showed better hydroisomerization selectivity (i/n ratio = 4.8) as compared to the previous studies in a single step process using single catalyst, which improve the freezing point of the fuel.

Kerosene range product distribution based on paraffins, naphthenes and aromatics (PNA) amounts for five different temperatures are compared in Fig. 2(d). When the reaction was carried out between 360 and 400 °C, the contribution of naphthenes and aromatics to the total products was negligible, as shown in Fig. 2(d). The product selectivity altered as the temperature was raised to 420 °C. The formation of paraffins (43.6 %), naphthenes (8.2 %) and aromatics (3.73 %) was found at high temperature (420 °C) and pressure (10 MPa). Partially unsaturated intermediates may have undergone cyclization proceeded by the dehydrogenation process to form highly persistent aromatics at higher temperature (420 °C). The product distribution assured the good fluidity and oxidative stability of the kerosene range HCs. The aromatic fraction of 18.7 % was obtained by Wang et al. [22] from catalytic hydrocracking of WCO. Furthermore, Wang et al. [28] produced jet range aromatics (7.6 %) using WCO as a feed. Cheng et al. [4] produced high aromatic content jet fuel (13.4 %) at 400 °C temperature using nickel-based mesoporous zeolite catalyst. They have used three types of zeolites loaded with nickel to meet jet range specifications. However, in this study, a lower amount of aromatics (3.73 %) was produced than [4,22,28] in one step method, which meets the jet fuel specifications as per the ASTM 1655 standard. A minimal aromatic content is required in aviation fuel as they do not burn as efficiently and clearly as alkanes. Aromatics have deleterious properties like low heat of combustion and poor combustion quality, leading to incomplete combustion and, consequently, soot production [31]. Herein, a very less amount of aromatics (3.73 %) was formed after the hydrocracking of WCO, which is advantageous because less burning of aromatics offers the potential to add value to fuel to minimize emission characteristics. A research study by Zhang et al. found that the lower aromatic content in jet fuel resulted in improved thermal oxidation stability [32]. In another study, Sinha et al. [33] eliminated the aromatics present in jet fuel via hydrogenation and converted it to cycloalkanes to increase thermal stability and energy density as well as minimize soot production. Furthermore, cycloalkane formation provides the same potential advantages as aromatics by allowing fuels to fulfil density requirements and perhaps giving the seal swelling capacity given by aromatics today. The jet fuel range HCs, which have cycloalkanes and isoalkanes, burns significantly cleaner and might boost energy density and specific energy, also fulfilling flash point, freezing point and O-ring swelling requirements [34].

Besides hydrocracking and isomerization reactions, heteroatom (nitrogen, sulfur and oxygen) removal is also a category of reaction that occurs during hydroprocessing reactions. Despite the fact that a significant amount of sulfur was intentionally introduced to the feedstock to modulate catalyst activity, the resulting kerosene HCs had just content of sulfur. The absolute values of N, S and O for feed and hydrocracking products at five different temperatures are given in Table 2. Fig. 2(e) compares the removal of heteroatoms as a percentage of their content in the feed at five temperatures during hydrocracking processes. The figure shows that nitrogen was the most successfully eliminated of the three elements, with a removal rate of over 98.3 % in all cases. The highest removal of nitrogen (99.8 %) was found at 420 °C. Sulfur was also successfully removed by over 91.5 % for all temperatures. However, the final product included a minimal quantity of sulfur (27–33 ppm), as seen in Table 2. Removal of sulfur gradually increased with increasing temperature. In particular, the sulfur removal rate increased from 91.5 % at 380 °C to 93 % at 440 °C. Furthermore, eliminating oxygen from the feed was the most difficult task. The amount of oxygen in the product was derived indirectly after the amount of C, H, N, and S were measured in the product, rather than by direct analytical testing. The lowest (97.5

Table 2
Comparison of elements present in feedstock and products at different hydrocracking temperatures, LHSVs, Pressures and H_2/oil ratios.

	C (wt %)	H (wt %)	O (wt %)	S (ppm)	N (ppm)
Temperature (°C)[a]					
360	85.80	13.91	0.286	33	5.4
380	85.42	14.53	0.047	30	0.44
400	85.11	14.84	0.046	30	0.71
420	84.60	15.38	0.017	29	0.61
440	84.54	15.35	0.107	27	1.14
LHSV (h^{-1})[b]					
0.5	84.38	15.61	0.0067	32	0.69
1	84.60	15.38	0.017	29	0.61
2	84.48	15.42	0.0973	26	0.89
3	84.82	14.95	0.227	22	5.21
5	84.74	14.92	0.336	24	9.89
Pressure (MPa)[c]					
8	84.46	15.47	0.067	28	0.92
9	84.52	15.31	0.165	43	1.14
10	84.60	15.38	0.017	29	0.61
11	84.41	15.42	0.167	24	0.29
H_2/oil ratio (NL_{gas}/L_{liquid})[d]					
1800	84.97	14.93	0.097	30	1.41
2000	84.68	15.28	0.037	28	0.84
2200	84.60	15.38	0.017	29	0.61
2400	84.27	15.54	0.186	35	0.74

Constant reaction condition for a: LHSV = 1 h^{-1}, P = 10 MPa, H_2/feed = 2200 NL_{gas}/L_{liquid}, b: T = 420 °C, P = 10 MPa, H_2/feed = 2200 NL_{gas}/L_{liquid}, c: T = 420 °C, LHSV = 1 h^{-1}, H_2/feed = 2200 NL_{gas}/L_{liquid}, d: T = 420 °C, LHSV = 1 h^{-1}, P = 10 MPa.

%) removal percentage of oxygen was found at 380 °C. A rise in the temperature from 360 to 420 °C leads to an increase in the oxygen removal percentage to 99.8 %, as shown in Fig. 2(e). As a result, higher temperatures favoured the reduction of all heteroatoms from the final products. Bezergianni et al. [27] performed catalytic hydrocracking of WCO, and the maximum removal of oxygen was found to be 96.29 % at 390 °C using a commercial hydrocracking catalyst. Moreover, in another study by Bezergianni et al. [17], they used three catalysts; hydrotreating, mild hydrocracking and a severe hydrocracking catalyst for the hydroprocessing of WCO. They found that oxygen was effectively removed nearly 95 % of the feed initial content. It is clear that the selection of the catalyst is a vital step in determining the quality of the final product. The catalyst used in this study is a better deoxygenation catalyst and able to remove the oxygen by over 99.8 % from feed (WCO), which is greater than the previous studies [17,27] and advantageous as triglyceride oxygenated molecules are unfavourable for hydrocarbon-based transportation fuels because they lower heating value and cause storage stability concerns.

One important study is provided in Fig. 2(f) by Van Krevelen diagram, which depicts the molar H/C and O/C ratios of the WCO feed and products obtained after hydrotreatment at various temperatures. The Van Krevelen diagram clearly shows a significant difference between feed and final products for both ratios (O/C and H/C). This was anticipated as WCO is higher in oxygen and lower in hydrogen than any other product. When compared to the feed value, O/C value of the product obtained at 360 °C (lowest temperature) was decreased by nearly 45 times, while the same ratio was decreased around 750 times at higher temperature (420 °C). Moreover, the H/C ratio, which is defined as the ratio of hydrogen to carbon atoms in a molecule, plays a significant role in determining the specific energy of the fuel. From Fig. 2(f), a relative increase in the H/C ratio was observed with temperature. Specifically, the H/ C ratio was increased nearly 1.1 times at 360 °C while 1.21 times at 420 °C as compared to the value present in the feed. Fuel with higher H/C ratios has higher specific energies. As a result, the catalyst accelerated the deoxygenation process in the form of CO_2 by reducing the O/C ratio and raising the H/C ratio.

3.2. Influence of LHSV on product distribution

LHSV, a significant operational factor in continuous hydroprocessing processes for controlling catalyst efficacy and also catalyst life expectancy, is defined as the residence time of the feed molecules on the catalyst surface while passing through the reactor. The basic premise of this investigation is the influence of LHSV on hydrocracking and hydroisomerization. The hydroprocessing of WCO was investigated at five various LHSVs (0.5, 1, 2, 3, and 5 h^{-1}) and at constant reaction parameters; T = 420 °C, P = 10 MPa, and H_2/oil = 2200 NL_{gas}/L_{liquid}.

The conversion and the selectivities of naphtha, kerosene and diesel at five different LHSVs are depicted in Fig. 3(a). It is evident from figure that as LHSV increased, conversion decreased from 99.9 (LHSV = 0.5 h^{-1}) to 99.2 % (LHSV = 5 h^{-1}). The drop in the conversion with greater LHSV was expected since higher LHSVs result in a shorter reaction time between catalyst and feed. Nevertheless, the conversion was above 96.6 % at all LHSVs. On the other hand, regarding the individual product selectivities, the maximum kerosene selectivity (55.6 %) was observed at 1 h^{-1}. When the LHSV increased from 1 to 5 h^{-1}, a drastic drop in kerosene selectivity was observed from 55.6 to 21.9 %. In the meantime, naphtha selectivity also decreased from 34.1 (1 h^{-1}) to 13.6 % (5 h^{-1}), while diesel selectivity increased from 6.6 to 57.4 %. Therefore, kerosene and naphtha selectivities were not promoted by LHSV, while LHSV favoured diesel selectivity. It is possible that rising LHSV decreases the reaction time, resulting in less cracking and, as a result, fewer lighter molecules being produced (kerosene and naphtha).

Isomer yield and i/n ratio at different LHSVs are compared in Fig. 3 (b). It is clear from the figure that low isomer yield (64.6 %) and i/n ratio (1.82) were obtained at the lowest LHSV (0.5 h^{-1}) as the product could not move out promptly and hampered the consecutive cracking and isomerization reactions on the catalyst surface. An increase in LHSV to 1 h^{-1} led to the rise in isomer yield and i/n ratio to 82.9 % and 4.8, respectively, showing that reactant and catalysts have adequate contact time for isomerization reactions. Further increase in the LHSV to 2 h^{-1}, a sharp reduction in isomer yield (59.6 %) and i/n ratio (1.47) was observed. It is possible that the comparatively high input flow rate of feed caused insufficient hydroisomerization reaction time, resulting in a dramatic fall in isomer yields and i/n ratio. Moreover, a rise in LHSV to 3 and 5 h^{-1}, 57.2 % and 55.7 % of isomer yield and 1.34 and 1.25 of i/n ratio was achieved, which is also in decreasing trend. An increase in the LHSV reduced the reaction time between the catalysts and feed, which reduced the hydroisomerization reaction rate and, hence, was responsible for reducing the yield of isomers and the i/n ratio. In a recent study, Wang et al. [26] produced jet range HCs with a maximum i/n ratio value of nearly 1.2 at 1 h^{-1} LHSV in a two-stage process. It is clear that products obtained in the current study showed better hydroisomerization selectivity (4.8 i/n ratio at 1 h^{-1}) in a single stage process as compared to earlier report [26].

The corresponding product distribution (PNA) of kerosene range HCs is compared for five different LHSVs in Fig. 3(c). The figure revealed a significant contribution of paraffins and cycloalkanes followed by aromatics for LHSVs below 2 h^{-1}. Specifically, increasing the LHSV from 0.5 to 1 h^{-1}, the amount of paraffins, naphthenes and aromatics increased from 35.8 to 43.6 %, 6.8 to 8.2 %, and 1.7 to 3.7 %, respectively. However, the contribution of naphthenes and aromatics was insignificant for LHSVs ranging from 3 to 5 h^{-1} as higher LHSVs suggest a shorter reaction time of the feed and catalyst within the reactor, resulting in a lower possibility of the production of cycloalkanes and aromatics.

Sulfur and nitrogen were efficiently eliminated by over 91.7 and 96.9 %, respectively for all cases, as shown in Fig. 3(d). Specifically,

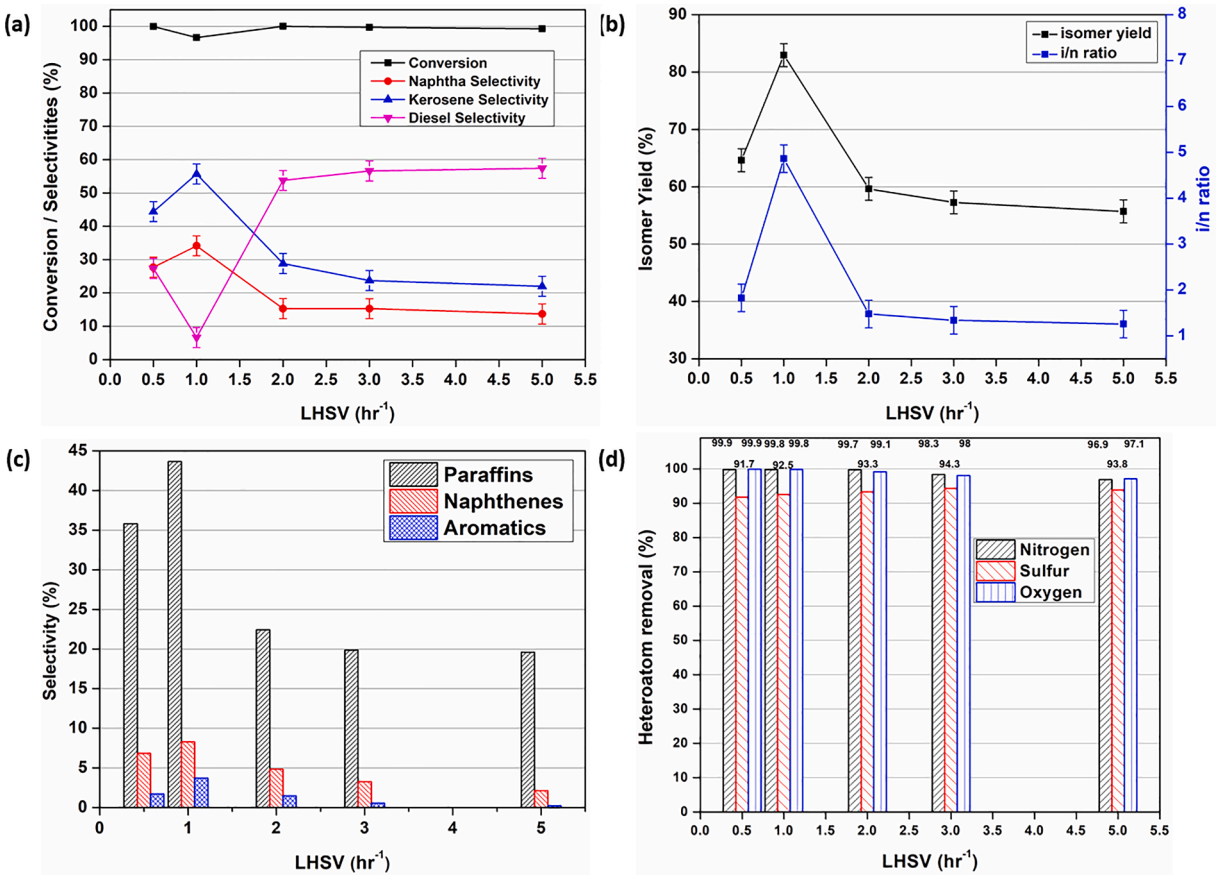


Fig. 3. Influence of LHSV on (a) conversion and selectivities, (b) isomer yield and i/n ratio, (c) PNA%, (d) heteroatoms (nitrogen, sulfur, oxygen) removal percentage.

maximum nitrogen was removed (99.8 %) at lower LHSV (1 h^{-1}). However, a nearly equal percentage of sulfur (91.7–94.3 %) was removed at all LHSVs. Moreover, oxygen was also effectively removed by over 97.1 % at all LHSVs. The oxygen removal rate was higher (more than 99.8 %) at lower LHSVs (0.5, 1 h^{-1}). Increase in the LHSV to 3 and 5 h^{-1}, the oxygen removal rate decreased to 98 and 97.1 %, respectively, due to a rise in the flow rate of feed, which lowers the contact time and, hence, a reduction in removal rate. To summarize, lower LHSVs exhibited the greatest removal of all three heteroatoms.

3.3. *Influence of pressure on product distribution*

Pressure is an essential factor because it influences the reactions such as hydrogenation, hydrocracking, and hydroisomerization. In this section, four experiments were carried out at 8, 9, 10, and 11 MPa. The other additional reaction conditions were; T = 420 °C, LHSV = 1 h^{-1}, and H_2/oil = 2200 NL_{gas}/L_{liquid}.

The conversion and product selectivities at different pressures are depicted in Fig. 4(a). It is evident from the figure that a nearly equal percentage of kerosene selectivity (38 %) was observed at 8 and 9 MPa pressure. When the pressure was raised from 9 to 10 MPa, the kerosene selectivity increased to 55.6 %. Moreover, the contribution of kerosene was significant at all pressures. Remarkably, kerosene exhibited the maximum selectivity (55.6 %) at 10 MPa pressure, while diesel production decreased with pressure due to more intense cracking. Mainly, the diesel selectivity decreased from 32.8 to 5.2 % as the pressure increased from 8 to 11 MPa, indicating more hydrocracking reactions at higher pressure. It is worth noting that kerosene emerged as one of the leading biofuels after the hydrocracking of WCO at all pressures.

Fig. 4(b) demonstrates the isomer yield and i/n ratio at four different hydrotreatment pressures. Isomer yield of 72.4 % and an i/n ratio of 2.6 was observed at the lowest pressure (8 MPa). Increase in the pressure from 8 to 9 MPa, isomer yield and i/n ratio increased to 75 % and 3, respectively. The maximum isomer yield (82.9 %) and i/n ratio (4.8) were observed when the pressure was raised to 10 MPa. It could be due to the fact that high pressure was responsible for hydroisomerization, which led to a high i/n ratio.

The product distribution for kerosene range HCs based on PNA selectivity at four different pressures was compared in Fig. 4(c). It is evident from the figure that 29.7 % of paraffins followed by 4 % of naphthenes and 4.9 % of aromatic HC selectivity, were observed at the lowest hydrotreatment pressure (8 MPa). Highest aromatics found at the same pressure as low pressure favours the formation of aromatics as it promotes cycloalkane dehydrogenation. Rise in the pressure from 8 to 10 MPa, the amount of paraffins (43.6 %) and naphthenes (8.2 %) increased, while aromatics (3.7 %) decreased.

The concentration of nitrogen, sulfur and oxygen reduced from the feed during the hydroprocessing was used to determine the percentage of heteroatom removal.

Nitrogen was most efficiently removed by over 99.6 % at all pressures, as shown in Fig. 4(d). Sulfur was also successfully eliminated in almost 88.9 % of all conditions. Furthermore, the most challenging molecule to eliminate was oxygen, which was effectively eliminated in all operating conditions by over 98.5 %. In particular, maximum oxygen was removed (99.8 %) at 10 MPa pressure. It is evident from the figure that all the pressures favour the removal of all the heteroatoms from the final products.

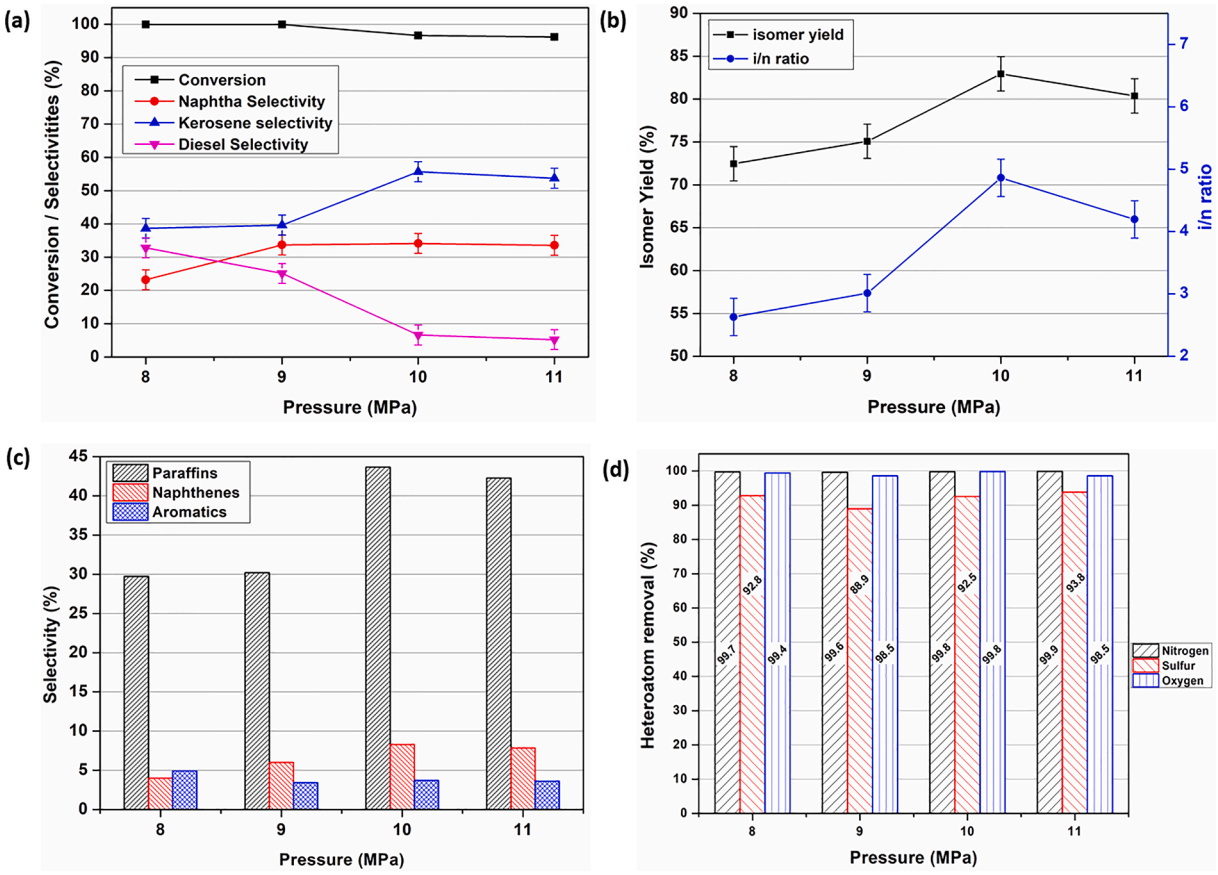


Fig. 4. Influence of pressure on (a) conversion and selectivities, (b) isomer yield and i/n ratio, (c) PNA%, (d) heteroatoms (nitrogen, sulfur, oxygen) removal percentage.

3.4. *Influence of H_2/oil ratio on product distribution*

An essential operational factor in hydroprocessing is the H_2/oil ratio since it is believed to have the greatest influence on the efficiency of hydrocracking and hydrogenation reactions. The H_2/oil ratio, in this work, refers to the hydrogen supply over WCO feed. In this section, four experiments were conducted at four H_2/oil ratios (1800 to 2400 NL_{gas}/L_{liquid}). The rest parameters were; T = 420 °C, P = 10 MPa, LHSV = $1hr^{-1}$.

The conversion and product selectivity at four different H_2/oil ratios were compared in Fig. 5(a). The results indicated that the maximum WCO conversion (99.9 %) was observed at a lower H_2/oil ratio (1800 and 2000 NL_{gas}/L_{liquid}). However, when the H_2/oil ratio increased, WCO conversion efficiency declined, which was predicted since unwanted cracking and hydrogenation reactions may occur along with favourable hydroprocessing reactions. Moreover, kerosene selectivity was favoured by increasing the H_2/oil ratio, as shown in Fig. 5(a). Kerosene selectivity risen from 40.6 to 55.6 %, while diesel selectivity decreased from 25 to 6.6 % when the H_2/oil ratio increased from 2000 to 2200 NL_{gas}/L_{liquid}. It might be due to the fact that heavy and diesel range HCs convert into lower range HCs, particularly in kerosene and naphtha range HCs. Further, an increase in the H_2/oil ratio to 2400 NL_{gas}/L_{liquid} did not favour kerosene range formation. High hydrogen pressure and rise in H_2/oil ratio may cause a diffusion constraint of reactants and intermediates. It is important to note that more than 40 % kerosene selectivity was observed at all H_2/oil ratios, indicating that 40 % of the WCO has transformed into kerosene at all H_2/oil ratios.

Isomer yield and i/n ratio at four H_2/oil ratios were compared in Fig. 5(b). The results showed that increasing the H_2/oil ratio enhanced isomer yield and i/n ratio, and the highest isomer yield (82.9 %) was found at 2200 NL_{gas}/L_{liquid} H_2/oil ratio. The high H_2/oil ratio enhanced the reaction rate of isomerization reactions, which improved the isomer production. Further increase in H_2/oil ratio from 2200 to 2400 NL_{gas}/L_{liquid} lowered the residence time, resulting in an inefficient reaction due to which the reactants were unable to change into isomers. It is worth noting that H_2 is an energy source; therefore, it is aimed at an operating state that uses the least amount of hydrogen possible.

Product distribution based on PNA selectivity of kerosene range HCs at four different H_2/oil ratios was compared in Fig. 5(c). It was evident from the figure that 28.5 % of paraffins, 1.5 % of naphthenes, and 4.8 % of aromatics (maximum) were found at the lowest H_2/oil ratio (1800 NL_{gas}/L_{liquid}). Low H_2/oil ratios promote the dehydrogenation of alkanes and cycloalkanes, leading to aromatics formation. Further increase in the H_2/oil ratio to 2200 NL_{gas}/L_{liquid} decreased the aromatics to 3.7 % while increasing the formation of paraffins and naphthenes to 43.6 and 8.29 %, respectively.

Fig. 5(d) compares the removal percentage of heteroatoms as a percent of their content in the feed at four H_2/oil ratios. Table 2 shows the amount of heteroatoms in the finished product. Nitrogen was the most successful element removed with over 99.5 %, whereas sulfur was similarly eliminated with over 91 %, as illustrated in Fig. 5(d). Furthermore, oxygen was also successfully removed by over 98.3 % at all H_2/oil ratios. Specifically, 99.1 % of oxygen was removed at the lowest H_2/oil ratio (1800 NL_{gas}/L_{liquid}). An increase in the H_2/oil to 2000 and 2200 NL_{gas}/L_{liquid} showed an increase in the oxygen removal rate to 99.6 and 99.8 %, respectively. Further rise in H_2/oil ratio showed a decrease in the oxygen removal rate to 98.3 %.

3.5. *Comparative study of the non-edible oils hydroprocessing and kerosene yield/selectivity*

In hydroprocessing experiments, a variety of catalysts, edible/non-

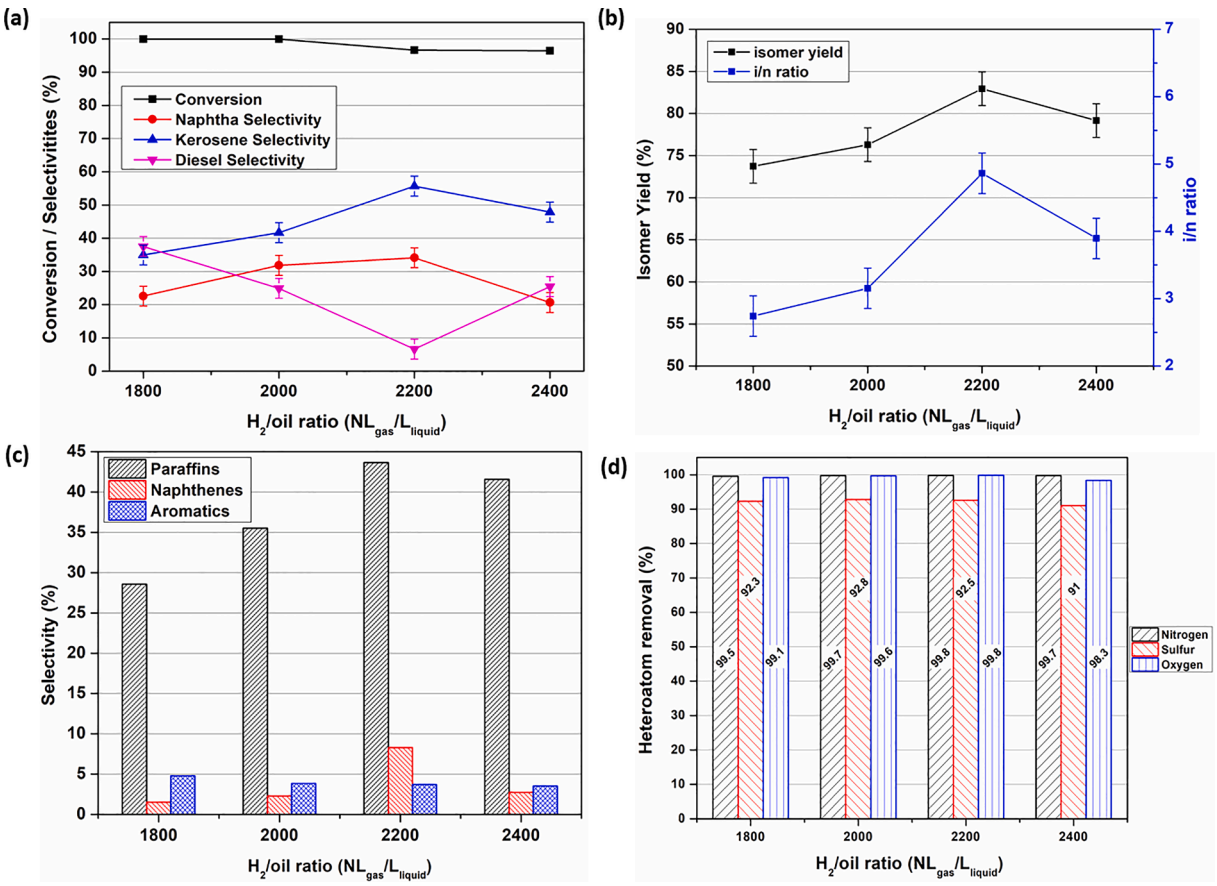


Fig. 5. Influence of H_2/oil ratio on (a) conversion and selectivities, (b) isomer yield and i/n ratio, (c) PNA%, (d) heteroatoms (nitrogen, sulfur, oxygen) removal percentage.

edible oils, and operating parameters are reported in the literature. The currently examined feed and catalyst results were compared to those of previously published systems, as shown in Table 4. Herskowitz et al. [35] have used soybean vegetable oil as a feedstock to produce jet and diesel HCs. The proposed process has three stages, and two different yields of biojet were achieved. The first stage involved loading soybean oil into a fixed bed reactor and performing hydrotreatment reactions at T = 370–385 °C, LHSV = 1.0 h^{-1} and H_2/oil = 800 NL/L. The liquid obtained from the first stage was further sent to two fixed bed reactors at 315 and 350 °C, respectively, with the stable operating condition; P = 3 MPa, H_2 flow rate = 300 NL/L, and LHSV = 4.0 $h^{-1,}$ and finally, 43.2 wt % bio-jet yield was produced with 12.5 % of aromatics. Moreover, another different method was adopted to produce biojet yield in which they fractionated the product into naphtha, jet and heavy. 12 wt% of biojet yield was produced. Then, hydrocracking of heavy fraction was performed to obtain 37 wt% jet yield. The combination of both jet yields was further isomerized, and finally, 48 wt% jet fuel and 14.3 % aromatics were achieved. The main disadvantage of this study is that they used multiple reactors and catalysts to get the maximum yield of jet fuel. Also, the feed used for hydroprocessing reactions is edible oil and can be used in the human diet. In a recent study, Wang et al. [26,30] synthesized biojet fuel from the hydroprocessing of waste cooking oil (WCO). The process involves two stages for the production of jet fuel. In the first stage, hydroprocessing of WCO has been conducted using two catalysts to achieve the alkanes ranging from C15 to C18, which are further sent to another reactor to match the jet fuel standards. In the second stage, hydrocracking and hydroisomerization of the mixture of *n*-alkanes (obtained in the first stage) took place to optimize reaction conditions and jet fuel standard specifications. This process requires multiple catalysts and reactors to achieve the maximum biojet yield. In another study, Cheng et al. [4] used WCO as a feedstock and produced jet fuel with 53 % alkane selectivity and 13.4 % aromatic HC selectivity. The produced fuel has a higher aromatic content resulting in higher particulate emission and lowers specific energy. Biojet fuel was synthesized from WCO by Wang et al. [22] in a continuous flow fixed bed reactor, and 39.7 % of biojet yield was obtained with an i/n ratio of 2.7. In this work, all processes like hydrogenation, cracking, and isomerization were performed concurrently in a single reactor and using a catalyst with strong hydrogenation/dehydrogenation ability along with desirable cracking function. Therefore, we have selectively produced aviation kerosene (55.6 % selectivity) with 8.2 % of naphthenes and 3.7 % of aromatics in a single step process using a low-cost feedstock (WCO). One of the potential reasons for the generation of naphthenes and aromatics in the current method might be the high exothermicity seen during these concurrent reactions in a single reactor [36].

3.6. *Catalyst lifetime (TOS)*

Long term runs of 590 h were carried out at T = 420 °C, LHSV = 1hr^{-1}, P = 10 MPa, and H_2/oil ratio = 2200 NL_{gas}/L_{liquid} to study catalyst lifetime and the influence of TOS, as shown in Fig. 6. The results reveal that the activity of the product did not decrease with time, and there was no noticeable change in the colour of the product. In particular, the constant selectivity (96.1–96.4 wt%) of liquid HCs (C5 - C18) was noticed until 220 h of continuous operation. Moreover, insignificant change in conversion (96.5–96.6 %) and > C18 (oxygenated by-products, 0.2–0.4 wt%) was observed until 220 h. During the catalytic hydrocracking process, >C18 products increased to 7.8 % until 590 h run, implying that the generation of these undesired products was controlled even for a lengthy run in a fixed bed reactor. In addition, the

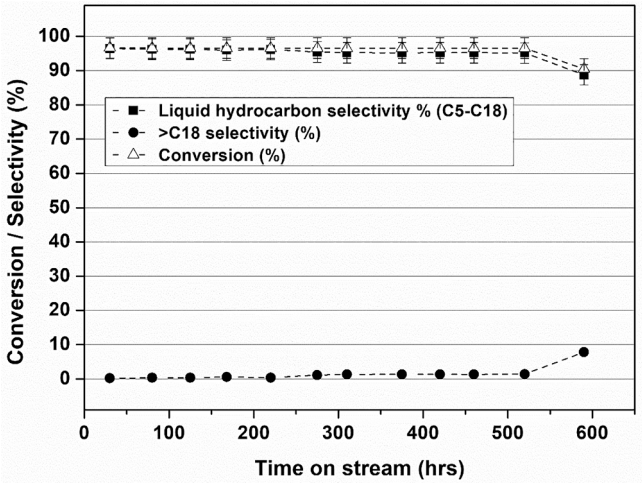


Fig. 6. Catalyst evaluation as a function of TOS at T = 420 °C, LHSV = 1 h^{-1}, P = 10 MPa, H_2/oil ratio = 2200 NL_{gas}/L_{liquid}.

conversion and selectivity of C5 - C18 HCs decreased to 90.5 % and 88.7 % until 590 h. Thus, there was no indication of the catalyst deactivation over time until 520 h. The product complied with all ASTM D1655 for aviation kerosene Jet A-1 (Table 3).

The reason for catalyst deactivation was found by BET, TGA, HR-TEM, FE-SEM and FT-IR analysis. The BET isotherm was utilized to compare the physicochemical parameters of the fresh and used catalysts (Fig. S1). The surface area and pore volume of the used catalyst were measured after it had been tested for 590 h at 420 °C, 10 MPa pressure, 1 h^{-1} LHSV, and 2200 H_2/oil NL_{gas}/L_{liquid}. Due to coke formation on the active surface sites of the catalyst, the reduction of total pore volume from 0.4584 to 0.131 cm^3/g and surface area from 152 to 39.7 m^2/g, respectively, was noticed (Table S1). The weight loss due to the unreacted HC and organic molecules was clearly seen in the TGA/DTG analysis of the used catalyst (Fig. S2). The morphology of the used catalyst by HR-TEM analysis also confirmed the presence of residual particles and carbon coking over the catalyst surface (Fig. S3). The FE-SEM analysis of the spent catalyst clearly depicts the presence of coke aggregation (Fig. S4). The IR spectra of the fresh and used catalyst were compared in Fig. S5. The IR spectra of the used catalyst clearly depict the presence of organic molecules, alicyclic HC and aldehydes, which lead to the deactivation of the catalyst and decrease the lifetime. Pre-treating the feedstock and employing suitable guard beds at the top of the reactor to minimize organic molecules and metal deposition over the catalyst surface are required to extend catalyst aging and enhance process economics.

Table 3
Properties of distilled kerosene HCs (C9 - C15) obtained at 420 °C temperature, 1 h^{-1} LHSV, 10 MPa pressure, 2200 NL_{gas}/L_{liquid} H_2/oil ratio as per ASTM D1655/D7566 [24].

Properties	Method	Limits	Result
Freezing point (°C)	D 5972	−40 Jet A −47 Jet A-1 (max)	−54.2
Density (kg m^{-3} at 15 °C)	D 4052	775–840	783.7
Flash point (°C)	D 3828F	38 (min)	42.8
Smoke point (mm)	D 1322	18 (min)	25
Net heat of combustion (MJ kg^{-1})	D 4529	42.8 (min)	44.8
Aromatics (vol%)	D1319	25 (max)	3.73
Distillation temperature (°C)	D 86D		137.4
10 % recovered (°C)		205 (max)	166.1
50 % recovered (°C)		Report	196.4
90 % recovered (°C)		Report	246.7
Final boiling point (°C)		300 (max)	258
Distillation residue (%)		1.5 (max)	0.9
Distillation loss (%)		1.5 (max)	0.3
Acidity (mg KOH g^{-1})	D 3242	0.1 (max)	0.0032
Viscosity (mm^2s^{-1} at −20 °C)	D 445	8 (max)	3.57
Gum content (mg 100 ml^{-1})	D 381	7 (max)	4.8
Sulfur (wt%)	D 4294	0.30 (max)	0.0029

3.7. *Plausible hydrocracking mechanism for converting WCO to kerosene range HCs*

We proposed a plausible mechanism for converting WCO to kerosene range HCs via the catalytic hydroprocessing route. Hydroprocessing of glycerides involves the hydrogenation of double bonds (unsaturation) within the side chains [37] and the removal of oxygen at the metal sulfide site [38]. The fatty acids present in the feed containing double bonds in the chain included palmitoleic acid (C16:1), oleic acid (C18:1) and linoleic acid (C18:2), as shown in Table 1. The double bonds present in the carbon chain of triglycerides are first saturated via the hydrogenation process in the presence of the catalyst at 420 °C and 10 MPa pressure. The propane backbone is then removed from the hydrogenated glycerides, which form free fatty acids (FFA). The produced FFAs are then converted into straight chain alkanes through the deoxygenation process, which includes hydrodeoxygenation (HDO), decarbonylation (DCO), and decarboxylation (DCO_2) reactions [37], leading to the formation of the octadecane compound [39]. A certain level of hydrocracking/isomerization is accompanied by deoxygenation at high temperatures and pressure to form naphtha, kerosene and diesel HC range compounds [40–43]. Alkanes underwent dehydrogenation, which resulted in the formation of alkene molecules. These alkene compounds cyclized through the carbenium ion and formed the tertiary cyclic carbocation. These cyclic carbocations are further converted into two types of compounds; one is substituted naphthenes, and another one is substituted aromatics [40,43].

4. Conclusions

WCO hydrocracking is a promising method for producing biofuels. In this work, WCO as a feed was hydroprocessed in a fixed bed reactor to yield high-quality kerosene range HCs. The experimental results showed that the higher selectivity of kerosene (55.6 %) was achieved at a temperature of 420 °C, LHSV 1 h^{-1}, pressure 10 MPa, and H_2/oil ratio 2200 NL_{gas}/L_{liquid}. Furthermore, high isomer yield (82.9 %) and i/n ratio (4.8) were achieved at the optimized reaction condition. The majority of heteroatoms were also removed (N = 99.9 %, S = 94.3 %, O = 99.9 %) from WCO during catalytic hydrocracking. Kerosene range HC selectivity based on paraffins, naphthenes, and aromatic was found to be 43.6 %, 8.2 %, and 3.73 %, respectively. The final product meets all the properties as required by the ASTM D1655 standard. The catalyst life in terms of conversion and total liquid HCs selectivity was carried out, and the results show no deactivation of the catalyst up to 520 h, confirming the high stability of the catalyst with time.

CRediT authorship contribution statement

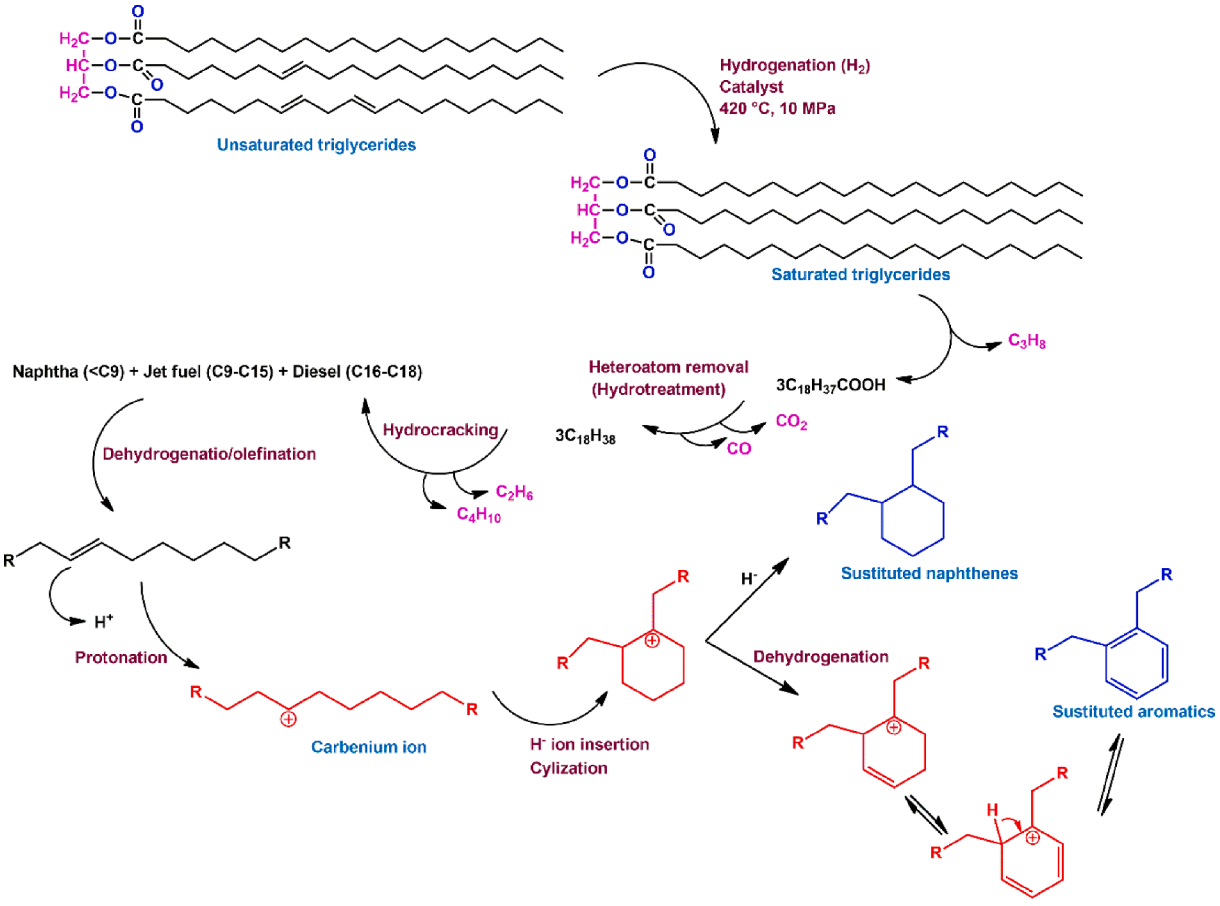
Vikas Verma: Conceptualization, Methodology, Validation, Investigation, Resources, Writing - original draft, Writing - review & editing. **Ankit Mishra:** Visualization, Writing - review & editing. **Mohit Anand:** Writing - review & editing. **Saleem Akhtar Farooqui:** Writing - review & editing. **Anil Kumar Sinha:** Supervision, Conceptualization, Methodology, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 4
A comparative study of the hydrocracked products obtained after the hydroprocessing of edible and non-edible oils.

Raw feed	Operating conditions				Reactor	Biojet Yield/Selectivity (%)	Ref
	T (°C)	P (MPa)	H_2/oil or H_2 flow rate	LHSV (hr^{-1}) or liquid flow rate			
Soybean oil	370–385 °C	3	800 NL/L	1	Fixed bed	48	[35]
WCO	320–400 °C	4	13.6 M ratio	2	Fixed bed	38	[26,30]
WCO	370–430 °C	3	–	–	Batch	53	[4]
WCO	350–400 °C	–	250 ml mL^{-1}	0.8 ml/min	Fixed bed	39.7	[22]
WCO	360–440 °C	10	2200 NL_{gas}/L_{liquid}	1	Fixed bed	55.6	This work



Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2022.126348.

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